

Supervised Learning – Regression

What is Regression?

Regression is a type of **Supervised Learning** where:

- The **output (Y)** is a **continuous value (numerical)**.
- The model predicts a **quantity** based on **input features (X)**.

Example:

 Predicting house prices based on size, location, and number of rooms.

Difference Between Classification and Regression

Task Type	Output	Example
Classification	Categories / Classes	Spam or Not Spam, Low/Medium/High
Regression	Continuous Values (Numbers)	Predicting house price, temperature

Key Terminology

Term	Meaning
Features (X)	Input variables or attributes.
Target (Y)	Continuous output value to predict.
Training Data	Labeled data used to train the model.
Testing Data	Data used to evaluate model performance.
Regressor	The algorithm/model used for regression tasks.

How Regression Works – Workflow

1.  **Collect labeled data** (input → continuous output pairs).
2.  **Select a regression algorithm** (e.g., Linear Regression).
3.  **Train the model** to learn the relationship between X and Y.
4.  **Test the model** on unseen data.
5.  **Evaluate performance** using appropriate metrics.

Common Regression Algorithms

Algorithm	Suitable For
Linear Regression	Simple, linear relationships.

Algorithm	Suitable For
Polynomial Regression	Curved or non-linear relationships.
Decision Trees	Complex relationships without assuming linearity.
Random Forest Regression	Large, complex datasets.
Support Vector Regression (SVR)	High-dimensional data.
Neural Networks	Very complex, non-linear patterns.

Example: Predicting House Prices

Feature (X)	Label (Y)
Size (square feet), Location, Bedrooms	House Price (in PKR)

Task:
The model learns patterns from these features and predicts the price for new houses.

Evaluation Metrics (Key Performance Indicators)

Metric	Meaning
Mean Squared Error (MSE)	Measures average squared difference between predicted and actual values.
Root Mean Squared Error (RMSE)	Square root of MSE, gives error in original units.
Mean Absolute Error (MAE)	Average of absolute differences between predicted and actual values.
R ² Score (Coefficient of Determination)	Indicates how well the model explains the variance in data (ranges from 0 to 1).

Real-World Examples

-  **House Price Prediction** → Predicting property value.
-  **Stock Price Forecasting** → Predicting future stock prices.
-  **Weather Prediction** → Estimating future temperature or rainfall.
-  **Sales Forecasting** → Predicting future sales volume.

Quick Summary

Task Type	Output Type	Example
Regression	Continuous Number	Predicting house price, temperature, sales